Project Title:
Advanced Power BI Sales Analysis Dashboard (Online Retail Dataset)

Course:
Data Visualization and Business Intelligence (DSA3050A)

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Introduction

This project presents an interactive Power BI dashboard developed using the Online Retail dataset. The main objective is to analyze sales performance, customer behavior, and product trends to support data-driven decision-making. The report includes data acquisition, cleaning, modeling, and visualization processes. The final dashboard is designed using a star schema and includes multiple interactive pages such as executive summary, detailed analysis, and performance monitoring. The insights generated help identify key business patterns, opportunities, and areas for improvement.

Part A: Data Acquisition and Understanding

1. Domain and Business Problem Being Solved

Domain:
Retail / E-commerce Sales Analytics

Business Problem:
An online retail store wants to analyze its sales transactions to understand customer purchasing behavior, product performance, and sales trends over time. The goal is to identify top-selling products, analyze customer distribution, and track revenue trends to support better inventory planning and business decision-making.

2. Dataset Source

The dataset used is the **Online Retail Dataset**, obtained from the UCI Machine Learning Repository. https://archive.ics.uci.edu/dataset/352/online-retail

Dataset Details:

- **Dataset Name:** Online Retail Dataset
- **Format:** Excel / CSV
- **Number of Records:** 541,909 transactions
- **Time Period:** December 2010 – December 2011
- **Data Type:** Transactional retail sales data

The dataset contains real-world online retail transactions including product sales, customer information, quantities, prices, and purchase dates.

3. Key Tables and Fields

Although the dataset is provided as a single table, it is structured into logical tables in Power BI for analysis.

Fact Table: Transactions (Sales)

This is the main transactional table. **Fields:**

- InvoiceNo — Order ID
- StockCode — Product ID
- Description — Product name
- Quantity — Number of items purchased
- InvoiceDate — Date of transaction
- UnitPrice — Price per unit
- CustomerID — Customer identifier

- Country — Customer location

Dimension Table: Customers

Created from customer-related fields. **Fields:**

- CustomerID — Primary key
- Country — Customer country

Dimension Table: Products

Created from product-related fields. **Fields:**

- StockCode — Product ID
- Description — Product name

Dimension Table: Date

Created from InvoiceDate. **Fields:**

- Date
- Year
- Month
- Quarter

Relationships

- Customers → Transactions (CustomerID)
- Products → Transactions (StockCode)
- Date → Transactions (InvoiceDate)

4. Why the Dataset Is Suitable for Advanced Power BI Analysis

This dataset is suitable because:

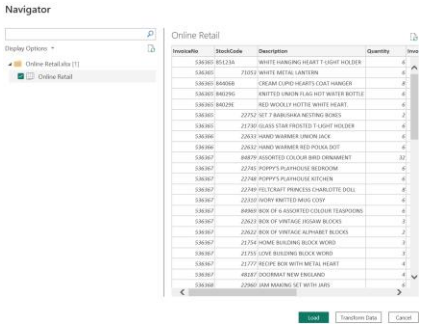
- It contains **over 500,000 records**, supporting large-scale analysis
- It includes a **date field (InvoiceDate)** for time intelligence analysis
- It has both **categorical variables** (Country, Description) and **numerical variables** (Quantity, UnitPrice)
- It supports creation of **multiple related tables** for relational modeling
- It enables advanced analysis such as:
    - Sales trends over time
    - Geographic sales distribution
    - Product performance analysis
    - Customer behavior analysis

Part B: Data Cleaning and Transformation in Power Query (Online Retail Dataset)

1.Import the dataset into Power BI

ıvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
536365	85123A	WHITE HANGING HE	6	12/1/2010 8:26	2.55	17850	United Kingdom
536365	71053	WHITE METAL LANT	6	12/1/2010 8:26	3.39	17850	United Kingdom
536365	84406B	CREAM CUPID HEAR	8	12/1/2010 8:26	2.75	17850	United Kingdom
536365	84029G	KNITTED UNION FLA	6	12/1/2010 8:26	3.39	17850	United Kingdom
536365	84029E	RED WOOLLY HOTTI	6	12/1/2010 8:26	3.39	17850	United Kingdom
536365	22752	SET 7 BABUSHKA NE	2	12/1/2010 8:26	7.65	17850	United Kingdom
536365	21730	GLASS STAR FROSTE	6	12/1/2010 8:26	4.25	17850	United Kingdom
536366	22633	HAND WARMER UN	6	12/1/2010 8:28	1.85	17850	United Kingdom

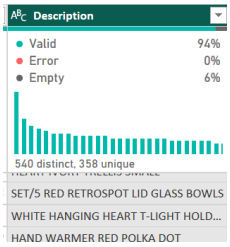
RAW DATA



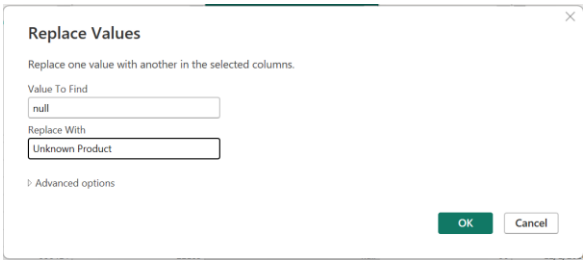
LOADING DATASET

2.Data Cleaning Process

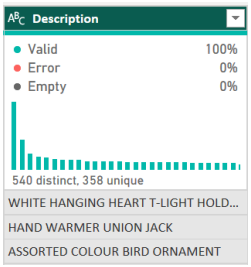
- Handling Missing Values



before



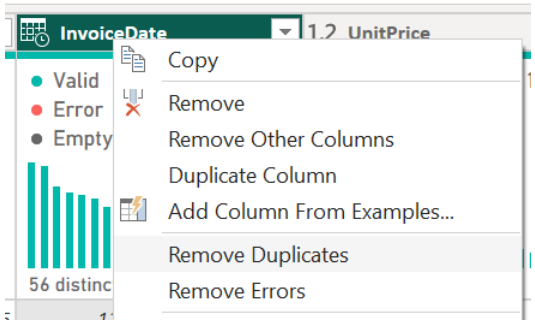
handling missing values



after

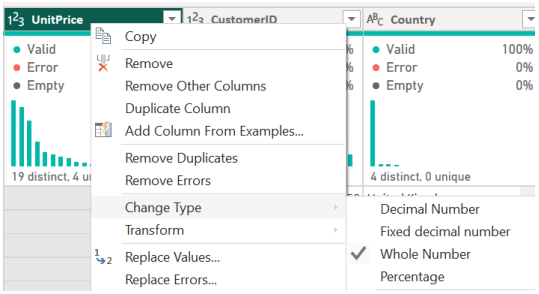
Reason: Missing identifiers affect relationship building and customer analysis accuracy.

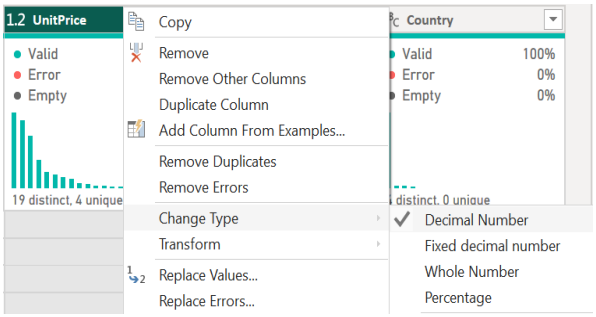
- Removing duplicates.



Reason: Prevents double counting of sales and ensures data accuracy.

- Correcting Data Types



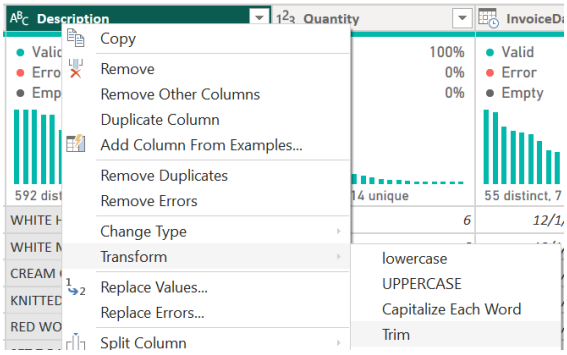


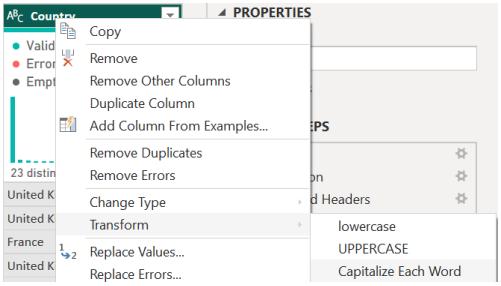
after

Before

Reason: Ensures correct calculations and time intelligence functions.

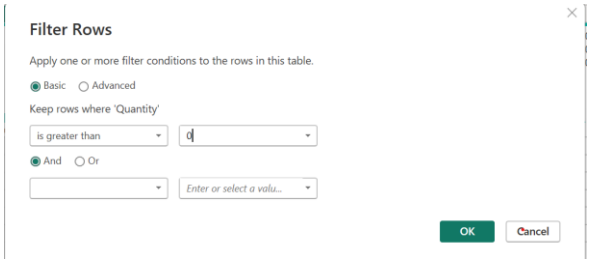
- Standardizing Text Values





Reason: Ensures consistent product categorization.

- Handling Invalid Records



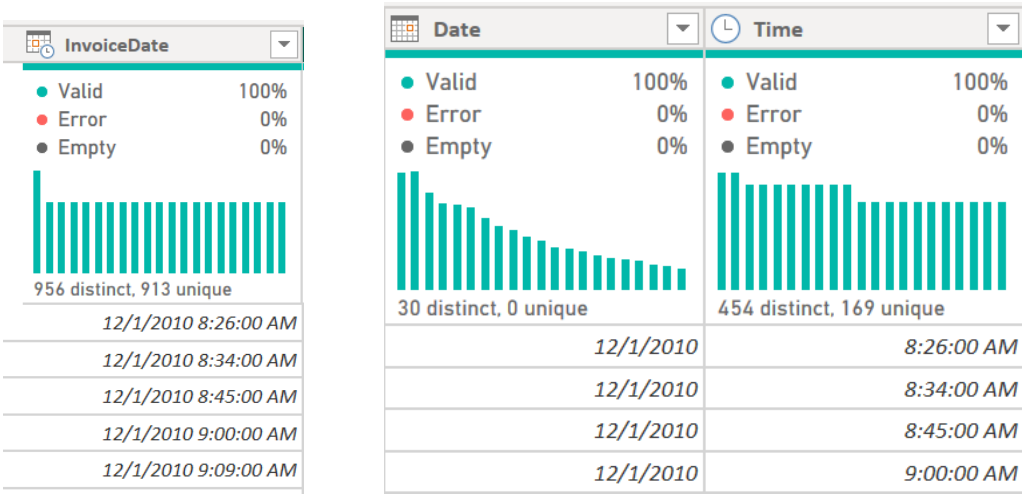
  - Removed records with:
    - Quantity ≤ 0
    - UnitPrice ≤ 0

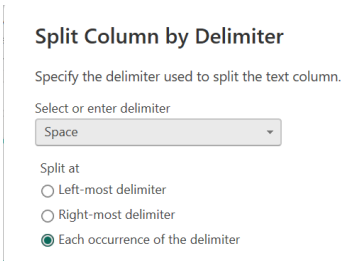
Reason: These represent returns or invalid transactions not useful for sales analysis.

3. Data Transformation Steps

1. Splitting Columns

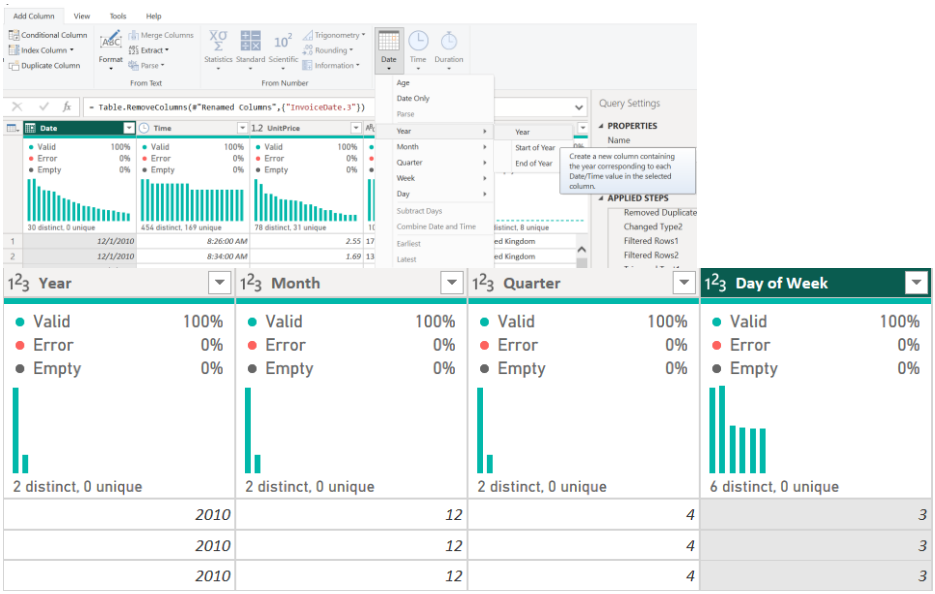
Split InvoiceDate into date and time





Purpose: Enables time-based analysis.

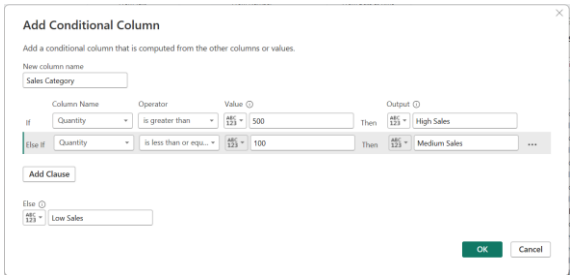
2.Extract Date Parts

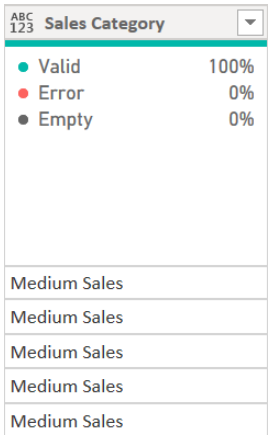


From Date: Year, Month, Quarter, Day of Week was extracted.

Purpose: Supports time intelligence in Power BI.

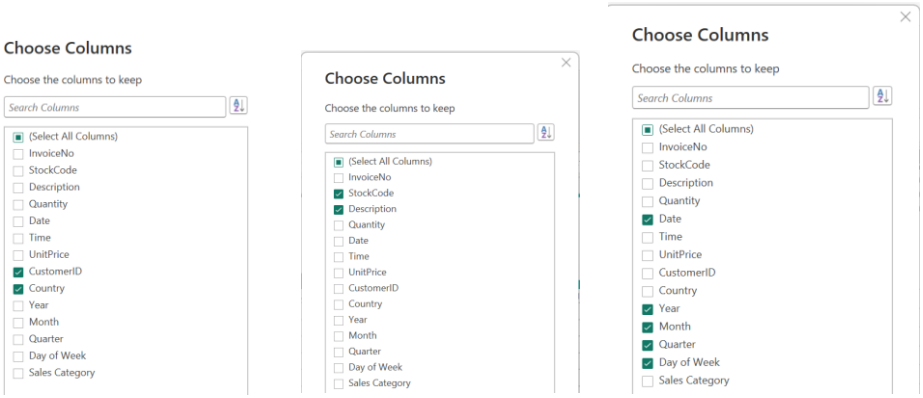
3. Creating Conditional Column





Purpose: Enables segmentation analysis.

4.Creating dimensional tables



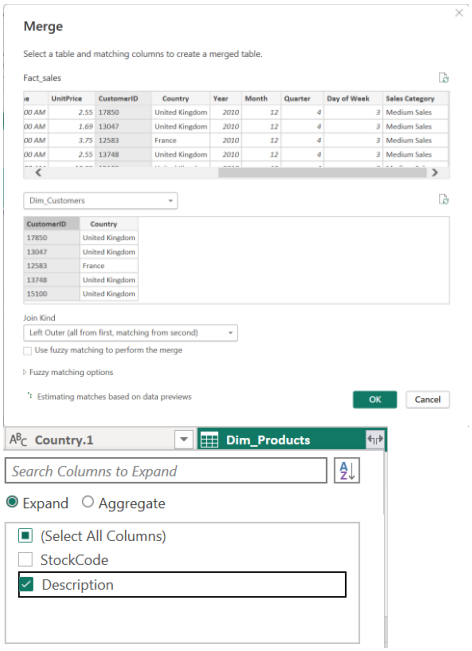
Created dimensional tables; Dim_customers, Dim_Products, Dim_Date

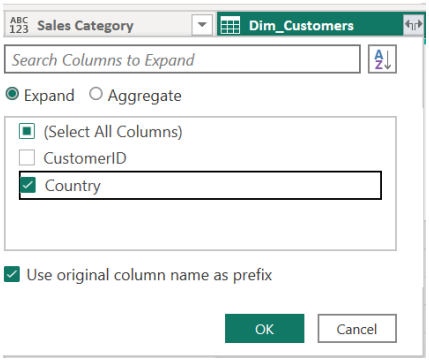
I created the dimension tables for Customers, Products, and Date by extracting the relevant fields from the main transaction table. I removed duplicates to ensure each record was unique and clean.

5. Merging Queries (Tables)

Merged:

- Fact_sales table with Customer dimension (CustomerID)
- Fact_sales table with Product dimension (StockCode)





Purpose: Builds relational structure

Part C: Data Modeling

1. Relationships Among Tables

The following relationships were created in Model View:

- **Fact_Sales[CustomerID] → Dim_customers[CustomerID]**
- **Fact_Sales[StockCode] → Dim_Products[StockCode]**
- **Fact_Sales[Date] → Dim_Date[Date]**

These relationships link the fact table to the dimension tables using key fields.

2. Fact and Dimension Tables

Fact Table

Fact_Sales

Fields include:

- InvoiceNo
- StockCode
- CustomerID
- Quantity
- UnitPrice
- Date
- Time

Justification:
Fact_Sales stores transactional data used for calculations such as total sales and quantity.

Dimension Tables

Dim_customers

- CustomerID
- Country

Dim_Products

- StockCode
- Description

Dim_Date

- Date
- Year
- Month
- Quarter
- Day
- Day Name

Justification:
Dimension tables store descriptive information used to filter and analyze sales data.

3. Cardinality and Cross-Filter Direction

For all relationships:

- **Cardinality:** Many-to-One (*:1)
- **Cross-filter direction:** Single

Justification:
This improves filtering performance and ensures efficient data processing.

4. Date Table Creation and Relationship

A Date table (**Dim_Date**) was created from the Date column.

Relationship created:

- **Fact_Sales[Date] → Dim_Date[Date]**

Justification:
The Date table supports time-based analysis such as monthly and yearly sales trends.

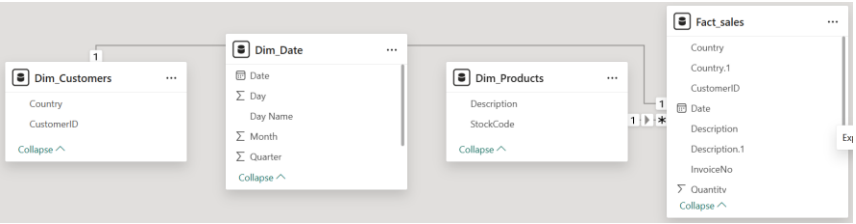
5. Why the Model Supports Efficient Reporting

The model follows a **star schema**, with one fact table connected to multiple dimension tables.

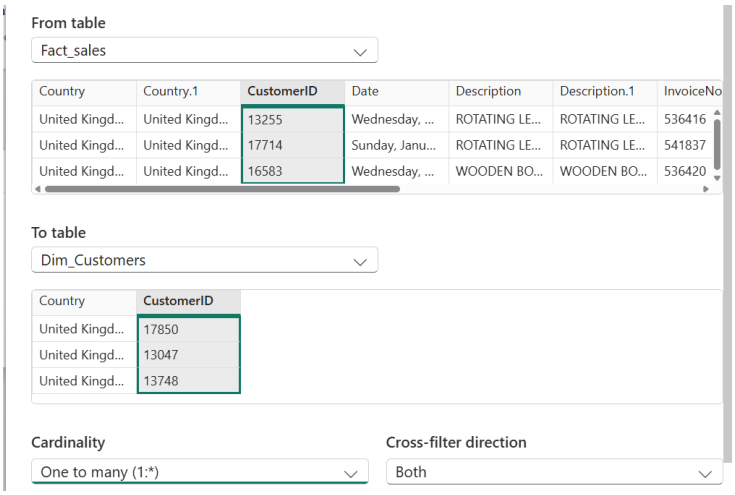
This structure:

- Improves performance
- Supports fast filtering
- Simplifies reporting
- Enables time intelligence analysis

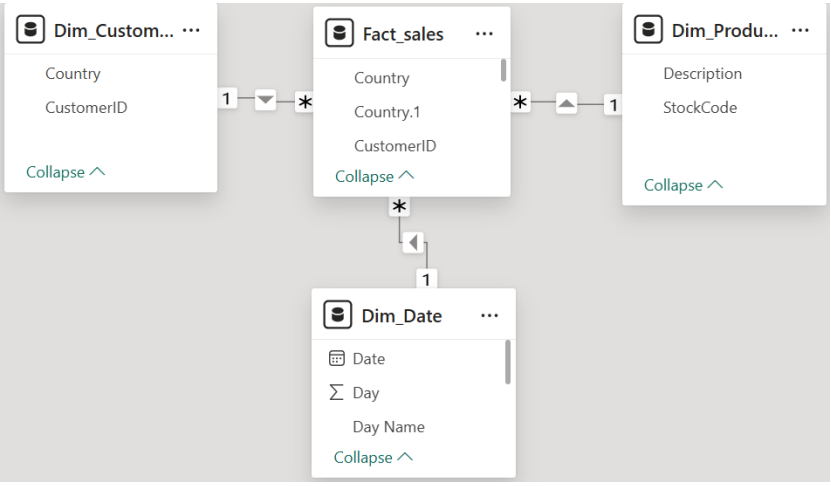
Before



Creating relationships



After



Part D: DAX Measures and Calculated Columns

Calculated Columns;

Calculated Column 1: Total Sales Value

Name: **`Sales Amount`**

```
Sales_Amount = Fact_Sales[Quantity] * Fact_Sales[UnitPrice]
```

This column calculates the total sales value for each transaction by multiplying Quantity by UnitPrice.

Calculated Column 2: Sales Category

Name: **`Sales_Category`**

```
Sales_Category =
IF(
    Fact_Sales[Sales_Amount] >= 500,
    "High Value Sale",
    IF(
        Fact_Sales[Sales_Amount] >= 100,
        "Medium Value Sale",
        "Low Value Sale"
    )
)
```

Explanation:
This column categorizes transactions into High or Low sales based on the quantity purchased.

DAX Measures;

1. Total Sales

```
Total Sales =
SUM(Fact_Sales[Sales_Amount])
```

Explanation: Calculates total revenue generated from all sales transactions.

2. Total Quantity Sold

```
Total Quantity =
SUM(Fact_Sales[Quantity])
```

Explanation: Calculates total number of items sold.

3. Average Sales Value

```
Average Sales =
AVERAGE(Fact_Sales[Sales_Amount])
```

Explanation: Calculates the average value of sales transactions.

4. Distinct Customers

```
Distinct Customers =
DISTINCTCOUNT(Fact_Sales[CustomerID])
```

Explanation: Counts the number of unique customers.

5. Total Orders

```
Total Orders =
DISTINCTCOUNT(Fact_Sales[InvoiceNo])
```

Explanation: Counts the total number of unique orders.

6. Year-To-Date Sale

```
YTD Sales =
TOTALYTD(
    [Total Sales],
    Dim_Date[Date]
)
```

Explanation: Calculates cumulative sales from the start of the year to the selected date.

7. Previous Year Sales

```
Previous Year Sales =
CALCULATE(
    [Total Sales],
    SAMEPERIODLASTYEAR(Dim_Date[Date])
)
```

Explanation: Calculates sales from the same period in the previous year.

8. Sales Growth %

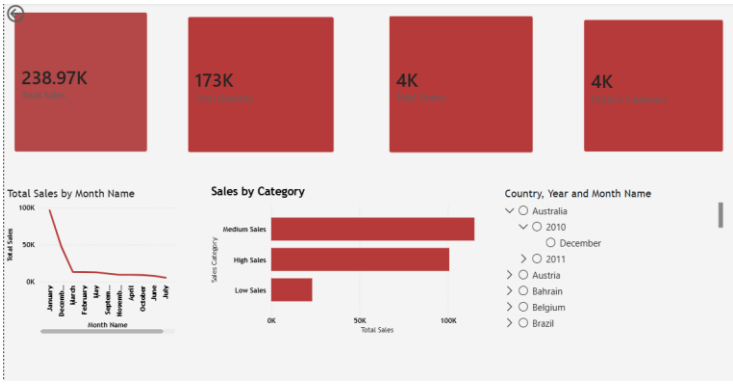
```
Sales Growth % =
DIVIDE(
    [Total Sales] - [Previous Year Sales],
    [Previous Year Sales]
)
```

Explanation: calculate % growth in sales compared to previous year.

Part E: Dashboard Design and Visualization

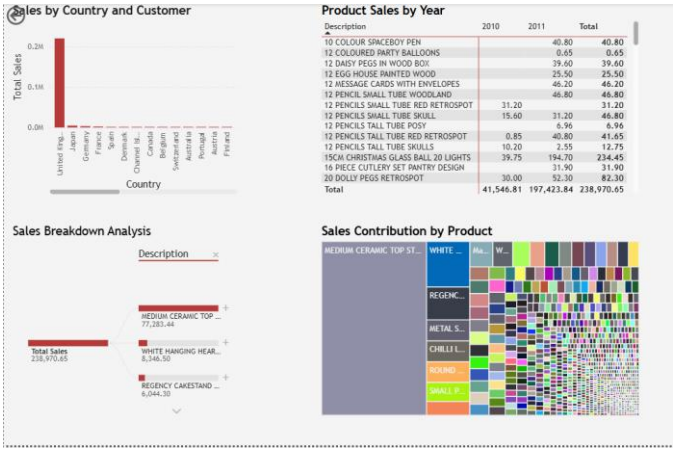
Page 1: Executive Summary

The Executive Summary page provides a high-level overview of business performance. It includes KPI cards for total sales, total quantity, total orders, and distinct customers. A line chart is used to show monthly sales trends over time. A bar chart compares sales across different categories. Slicers for country, year, and month allow users to filter the data interactively. The layout is simple, with KPIs at the top and charts below for clarity.



Page 2: Detailed Analysis

The Detailed Analysis page provides deeper insights into sales performance. A drill-down column chart is used to analyze sales from country to customer level. A matrix table shows product sales across different years. A decomposition tree is used to break down total sales by country, product, category, and customer. A treemap is included to show product contribution to total sales. This page focuses on detailed exploration and breakdown of data.



Page 3: Insights and Performance Monitoring

The Insights page focuses on performance tracking and comparison. A line chart is used for time intelligence analysis of sales trends. Bar charts are used to show top 10 and bottom 10 performing products. A clustered column chart compares current sales with previous year sales. A pie chart, donut chart, and map are used to show category distribution and geographic sales patterns. Slicers allow users to filter data by country, year, and month for interactive analysis

Design Choices

The dashboard uses a consistent theme, simple colors, and clear labels across all pages. Visuals are arranged in a logical structure starting with KPIs, followed by trends and detailed analysis. Only business-relevant visuals were used to ensure clarity and easy interpretation.

User Interaction

Users interact with the dashboard using slicers for country, year, and month. Cross-filtering allows visuals to update dynamically when a selection is made. Drill-down features allow users to explore data from summary to detailed levels. Tooltips provide additional information when hovering over visuals, making the dashboard interactive and easy to use.

Part F: Analytical Insights and Business Recommendations

Analytical Insights

1. Total sales show fluctuations over time, indicating unstable monthly performance rather than consistent growth.
2. A small number of products contribute a large share of total sales, showing high dependence on top-performing items.
3. There is significant variation in sales across different countries, indicating regional performance imbalance.
4. A small group of customers contributes most of the revenue, showing high-value customer concentration.
5. Certain product categories consistently underperform, suggesting weak demand or poor market fit.

Business Recommendations

1. The business should improve underperforming product categories by reviewing pricing, demand, and marketing strategies.
2. The company should reduce reliance on a few top-selling products by diversifying its product range.
3. Targeted marketing strategies should be implemented in low-performing regions to improve sales distribution.

Conclusion

The analysis identifies key patterns in sales performance, customer behavior, and product distribution. These insights support better decision-making, improved revenue stability, and overall business growth.

POWER BI LINK:
https://app.powerbi.com/view?r=eyJrIjoiZTIzOGQ1ZWEtMzA0YS00YzJkLTg4MmQtODkzYmIxZjQ2YWQ4IiwidCI6IjE2ZDgzZWU2LTI1NGEtNDY5ZC1hNmNjLTU0ZTJjYTIzMTNlNyIsImMiOjh9

GITHUB LINK: https://github.com/halima-04/DSA3050A-Advanced-PowerBI-Exam